

Application, services & system monitoring using temporal graph convolution network

As modern software architectures shift toward highly distributed microservices, traditional monitoring strategies—relying on static thresholds and isolated rule-based alerts—increasingly fail to capture the complex, cascading dependencies inherent in cloud-native environments. This project proposes a novel predictive monitoring framework leveraging Temporal Graph Convolutional Networks (T-GCNs) to model the evolution of system health in real-time.

By representing the infrastructure as a dynamic interaction graph, where nodes signify microservices or containers and edges represent network traces and API calls, the system effectively captures both spatial dependencies and temporal patterns. High-fidelity telemetry data, including CPU utilization, memory pressure, and request latency, are integrated as dynamic node features. The model architecture utilizes a hybrid approach—employing Graph Convolutional Networks (GCNs) to extract structural features and Long Short-Term Memory (LSTM) or Temporal Graph Networks (TGNs) to process the chronological sequence of system states.

The primary objective is to transition from reactive troubleshooting to proactive failure prediction, automated Root Cause Analysis (RCA), and early-warning anomaly detection. By learning how performance degradation propagates through the service mesh, this framework enables DevOps teams to identify bottlenecks and potential outages before they impact the end-user. Experimental results suggest that this graph-centric temporal approach significantly improves the accuracy of failure forecasting and reduces the Mean Time to Detection (MTTD) compared to traditional time-series analysis.